

Clinical Risk Assessment Report: Patient 37

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 12.8% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.58x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 3.9%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

- Squamous Cell Carcinoma Antigen (SCC_Ag)

Value: 0.6 [Bottom 18%] (+1.4%)

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.
- 75th Percentile Tumor Density (CONVENTIONAL_HUQ3)

Value: 0.69 [Avg (68th %ile)] (+1.2%)

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.
- Large Zone Emphasis (CT) (GLZLM_LGZE.CT)

Value: -0.26 [Avg (35th %ile)] (+0.9%)

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.
- Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT)

Value: -0.07 [Top 16%] (+0.7%)

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

3. Protective / Mitigating Factors (Reducing Risk)

- Small Zone Emphasis (PET) (GLZLM_SZE.PET)

Value: -0.17 [Avg (39th %ile)] (-3.9%)

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.
- Patient Age (Age)

Value: 66.0 [Avg (59th %ile)] (-1.9%)

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.
- Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT)

Value: 0.43 [Avg (65th %ile)] (-1.5%)

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.
- Maximum Tumor Density (CONVENTIONAL_HUmax)

Value: -0.1 [Avg (41th %ile)] (-1.3%)

Reflects the most dense solid components or calcifications within the primary tumor lesion.

Machine Learning Feature Attribution (SHAP Decision Path)

